Siddharth Desai

+91 98243 25333 | siddesai10032005@gmail.com | desaisiddharth.vercel.app

PROFILE

B.Tech ECE student passionate about Backend VLSI and Physical Design. Experienced with CMOS layouts, STA, and mixed-signal design. Skilled in FPGA systems, microcontrollers, and Verilog-based hardware development. Also pursuing a Minor in Marketing focusing on consumer behaviour, marketing management, digital marketing, and international marketing.

EDUCATION

Institute of Technology, Nirma University, Ahmedabad **2023 – 2027**
B.Tech in Electronics and Communication Engineering
CGPA: 7.60 (Overall till 5th SEM)

Bhulka Vihar School **2009 – 2023**
12th: 77.6% (95.8 Percentile)

SKILLS

Physical Design: Floorplanning, Placement & Routing, Timing Diagrams, STA (Setup/Hold)
CMOS Layout: Microwind, Magic, Inverter/Flip-Flop/Counter Layouts
Digital Design: FSMs, UART, Counters, Clocking, Metastability
Analog Circuits: Op-Amp Testing, Precision Rectifiers, Slew Rate Measurement
FPGA/HDL: Verilog, SystemVerilog, ModelSim, Vivado, Debugging
Marketing: Consumer Behaviour, Digital Marketing, Data Analytics

PROJECTS

Constrained Random Verification of a RISC-V ALU Using SystemVerilog
- Developed a comprehensive constrained random testbench in SystemVerilog to verify a RISC-V ALU.
- Implemented functional coverage groups and assertions to ensure correctness across edge cases and corner conditions.

FPGA Based Neural Network Acceleration for Human Computer Interaction (HCI) Application
- Designed and deployed a hardware-accelerated neural network on FPGA for real-time HCI applications.
- Optimized inference pipeline for low-latency gesture and input recognition using custom Verilog modules.

Line Follower and Object Detection Rover
- Built an autonomous rover with line-following logic and real-time object detection using sensor fusion.
- Integrated IR sensors and camera module with embedded firmware for navigation and obstacle avoidance.

SMPCache – Simulator for Cache Memory Systems on Symmetric Multiprocessors
- Simulated cache coherence protocols (MSI/MESI) on symmetric multiprocessor architectures.
- Analyzed performance metrics including hit/miss rates and coherence traffic under different workloads.

Precision Rectifier Using Op-Amp
- Designed a precision rectifier eliminating diode drop for accurate low-level rectification.
- Validated performance through simulation and hardware testing.

Digital Logic-Based Multi-Point Liquid Level Sensing System
- Developed a combinational-logic water-level detection system.
- Implemented discrete LED-based real-time indication.

UART Transmitter & Receiver on FPGA
- Built a Verilog UART Tx/Rx with baud-rate generator and frame validation.
- Verified through ModelSim simulation and terminal communication.

Slew Rate Testing of Practical Op-Amp
- Measured op-amp slew rate using oscilloscope-based step response.
- Compared theoretical vs practical performance.

8051 – Bluetooth-Based RGB Lighting & Auto Light Detection
- Developed an HC-05 Bluetooth-controlled RGB lighting system.
- Added automatic brightness control via proximity sensing.

CMOS Layout of 2-Bit Up/Down Counter (Microwind)
- Designed complete CMOS layout including routing and DRC fixes.
- Performed switching simulations for validation.

CMOS Temperature Sensor Using Ring Oscillator

- Built a temperature sensor based on ring oscillator frequency drift.
- Designed inverter-chain layout and performed calibration.

LEADERSHIP EXPERIENCE

Core Committee Member – Techno Cultural Committee (Vaudeville) 2026

- Managed disciplinary operations and coordinated major cultural events.

Executive Committee Member – Techno Cultural Committee (Vaudeville) 2024

- Organized drama events and supported university-wide cultural activities.

Core Committee Member – Techno Cultural Committee (Kala Carnival & Nuzeal) 2025–2026

- Assisted with planning, logistics, and execution of arts festivals.

Volunteer – Techno Cultural Committee (Vaudeville & Drama Events) 2023

- Contributed to smooth event execution during freshman year.